Sahaj Bhadja

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Summary

Software engineer with 2+ years developing **clean, modular code** using object-oriented principles. Strong foundations in **computer science fundamentals**, data storage technologies, DevOps/CI/CD, and test automation throughout the software development lifecycle.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Analysis of Algorithms, Data Structures, Database Systems, Complexity Analysis

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Object-Oriented Programming, Data Structures, Algorithms, Software Engineering

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Developed software products using **object-oriented design principles** in Python, writing code that is flexible, reusable, efficient, and adaptable to changing requirements while ensuring completion of tasks and quality of work.
- Built tools and produced **technical documentation** to improve developer efficiency, analyzing systems to determine properties and functions while supporting peers to ensure consistent team delivery.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed production software through the **software development lifecycle**, applying algorithm design and problem-solving skills to build features handling 10K+ daily requests with modular, maintainable code.
- Wrote comprehensive **unit tests and automated testing** using pytest, participating in code reviews and ensuring quality throughout development while adapting to changing business requirements.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Applied **algorithm design and complexity analysis** to optimize data processing pipelines, improving system performance by 18% through efficient data structures and problem-solving approaches.
- Built productive working relationships with cross-functional teams, sharing knowledge and presenting technical information while proactively seeking learning opportunities.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Designed and implemented software using **object-oriented development** in Python, building scalable systems with clean architecture that improved efficiency by **68%** through systematic analysis and optimization.
- Established **DevOps and CI/CD pipelines** using Jenkins, implementing test automation with pytest and ensuring code quality through comprehensive unit testing and code reviews within Agile workflows.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed software with **data storage technologies** including MongoDB and SQL databases, designing efficient schemas and queries to handle 2M+ records while applying software engineering best practices.
- Collaborated effectively with team members to resolve problems, recommending viable alternatives when faced with shifting priorities and ensuring actions aligned with team-level goals.

Skills

Languages: Python, Java, JavaScript, TypeScript, SQL (object-oriented development)
Data Storage: PostgreSQL, MySQL, MongoDB, database design, query optimization
DevOps/CI/CD: Jenkins, GitHub Actions, Docker, Git, automated deployment pipelines
Testing & Quality: Unit testing (pytest, Jest), test automation, code reviews, Agile/Scrum

Projects

Big-Brain Live | *Python, PostgreSQL, FAISS, Algorithm Design*

- Built scalable software applying **algorithm design and data structures** for efficient document retrieval across 50K+ items, implementing complexity-optimized search algorithms achieving 95% query accuracy.

Linguistix Code | *Python, Docker, AWS, Jenkins CI/CD*

- Developed containerized application with **DevOps and CI/CD pipelines**, implementing automated testing and deployment workflows with comprehensive unit tests and quality validation checks.

FinNexus Live | *TypeScript, MongoDB, REST APIs*

- Built financial platform using **object-oriented design** and data storage technologies, implementing secure transaction workflows with proper data validation for 50K+ users in regulated financial domain.